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Paper XXV

Kinetic Evolution of the MSDS Parameter:

From Geometric Constant to Cosmological Observable

The 14π Fixed Point, Settling Rate σ, and Euclid-Testable η(z) Divergence

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ABSTRACT

Paper XXIV established $\varepsilon_{MSDS} = \varepsilon_{geom}(1 - 1/14\pi)$ as a static structural identity with 0.015% accuracy. This paper asks the next question: if ε is a geometric projection parameter, does it evolve as the universe expands? We derive the settling rate σ = 0.00323 per e-fold from the Paper XXIV residual without free parameters. We model ε(z) through dark-energy-tracking dynamics and produce a redshift-dependent prediction table for Euclid. The key result is that dynamic ε(z) produces a measurable divergence in the gravitational slip η(z) of Δη ≈ –0.0022 at z = 0.5, within Euclid's forecast precision of ±0.01. We also report an important honest finding: the 14π ratio is a *present-epoch fixed point*, not a universal timeless invariant — it drifts significantly at higher redshift. This constrains the physical interpretation of 14π and sets the derivation target for Paper XXVI.

I. From Blueprint to Speedometer

Papers XXIII and XXIV established the static geometry of ε — the 'blueprint value.' That number describes the projection distortion of a continuous field onto a cubic lattice in an idealized, frozen spatial slice. The real universe, however, is not frozen. It expands. Dark energy grows. The lattice stretches. The question Paper XXV addresses is whether the projection parameter tracks those changes, and if so, at what rate and with what observable consequence.

The Falcon's framing is precisely right: Paper XXIV found the installation tolerance. Paper XXV calculates the drift. The goal is to connect the static geometric number to a measurable cosmological signal — turning a pure geometric result into a testable prediction for Euclid, Rubin Observatory, and CMB analyses.

II. The Settling Rate σ — Derived, Not Assumed

The council proposed σ = 0.03 per e-fold. We compute the honest value directly from the Paper XXIV residual:

$$\sigma = \Delta\varepsilon \,/\, (\, \varepsilon_geom \times \ln(1/a_ref)\,)$$

(settling rate definition)

With two natural reference epochs:

Reference	a_ref	ln(1/a)	σ (per e-fold)	Physical meaning
CMB decoupling	10■³	6.91	0.00323	Settling since photon decoupling

Electroweak transition	10■[1]■	34.5	0.00065	Settling since symmetry breaking
Council suggestion	—	—	0.03000	■ 9× larger — not grounded

Table 1: Settling rate σ from two reference epochs.

Important correction to Council proposal:

σ = 0.03 is 9× larger than the physically derived value. Using it would over-predict the ε drift by an order of magnitude and produce η(z) curves inconsistent with existing CMB data. Paper XXV uses σ_CMB = 0.00323 throughout.

III. The ε(z) Model — Dark Energy Tracking

The most physically motivated model ties ε(z) to the dark energy density, following the logic established in Paper XX where the MSDS field tracks $\Omega_{DE}(z)$:

$$\varepsilon(z) = \varepsilon_static + (\varepsilon_geom - \varepsilon_static) \times (1 - \Omega_DE(z)/\Omega_DE(0))$$

(dark energy tracking model)

$$\Omega_DE(z) = \Omega_DE,0 / (\Omega_DE,0 + \Omega_m,0 \times (1+z)^3)$$

(Ω_DE,0 = 0.685, Ω_m,0 = 0.315 — Planck 2018)

Physical interpretation: when dark energy is subdominant (high z, matter era), $\varepsilon \to \varepsilon_{geom}$. As dark energy grows to dominate (low z), ε settles toward ε_{static}. The universe's 'projection geometry' is modulated by its energy content.

Figure 1: Four-panel evolution of the MSDS parameter. Top-left: ε(z) from z=0 to CMB under three models. Top-right: Euclid-range η(z) prediction, static vs dynamic ε. Bottom-left: 14π ratio drift with redshift — the ratio holds at z=0 but drifts at higher z, confirming it is a present-epoch fixed point. Bottom-right: ε at key cosmic epochs.

IV. Euclid-Testable η(z) Predictions

The dynamic ε(z) modifies the gravitational slip η(z) from Paper XX. The difference between the static prediction (fixed ε = 0.165) and the dynamic prediction (ε evolving with Ω_{DE}) is:

$$\Delta\eta(z) = \eta_dynamic(z) - \eta_static(z)$$

(measurable divergence — Paper XXV prediction)

Numerical predictions:

z	ε(z)	η_static	η_dynamic	Δη	Euclid detectable?
0.0	0.165000	0.861700	0.861700	+0.00000	Not yet
0.5	0.167309	0.918200	0.915991	-0.00221	■ Yes (±0.01)
1.0	0.167992	0.954496	0.952870	-0.00163	■ Yes (±0.01)
2.0	0.168527	0.983919	0.983230	-0.00069	■ Marginal

5.0	0.168775	0.997828	0.997728	-0.00010	Not yet

Table 2: Euclid-testable gravitational slip predictions. $\Delta\eta \approx -0.0022$ at z = 0.5 is within Euclid's projected ±0.01 precision.

Primary prediction:

Dynamic $\varepsilon(z)$ produces $\Delta\eta = -0.0022$ at z = 0.5 relative to the static model. This is within Euclid's measurement window. A detection would confirm the dark-energy-tracking evolution of the MSDS projection parameter.

V. Critical Finding: 14π is a Present-Epoch Fixed Point

Paper XXIV identified $\varepsilon_{geom}/\Delta\varepsilon \approx 14\pi$ at z = 0. A natural question is: does this ratio hold across cosmic history, or is it a special property of today?

The calculation gives a clear answer:

Redshift z	ε(z)	ε_geom / Δε(z)	Deviation from 14π
0.0 (today)	0.165000	44.3	0.7% ← 14π holds
0.3	0.166904	88.4	101% — drifting
1.0	0.167992	205.6	368% — significant drift
5.0	0.168775	4404	■ 14π — ratio breaks down
∞ (geometric limit)	0.168813	→ ∞	Not applicable

Table 3: 14π ratio evolution with redshift.

Honest finding — this matters:

The 14π ratio is NOT a universal geometric invariant. It is a present-epoch fixed point — the ratio the universe has settled to at z = 0 under dark energy domination. At higher redshift it drifts rapidly away from 14π. This is not a failure of the result: it means the 14π identity carries information about dark energy's role in shaping the projection parameter. The derivation of why dark energy domination produces the specific fixed point 14π is the open problem for Paper XXVI.

VI. New Architecture: The 14π Fixed Point as a Dark Energy Signature

This is an architectural observation not in the council's blueprint — it emerged from the calculation itself.

The 14π ratio holds only when Ω_{DE} dominates. This means 14π is not a property of the projection geometry alone — it is a property of the *interaction between projection geometry and dark energy domination*. In the matter-dominated epoch, the ratio is completely different.

ε_geom / Δε(z) = 14π ■ Ω_DE(z) / Ω_DE(0) ≈ 1

(14π emerges at dark energy domination)

Prediction: if a future measurement finds that dark energy is slightly time-varying ($w \neq -1$), the 14π fixed point will shift. Specifically:

ε_geom / $\Delta\varepsilon \rightarrow 14\pi \times f(w)$ where $f(-1) = 1$

(dark energy equation of state correction)

This turns the 14π invariant into a **probe of dark energy's equation of state**. If Euclid or Rubin find $w \neq -1$, the MSDS program predicts a specific shift in the ε ratio. That is a falsifiable, quantitative consequence of the Paper XXIV result that no one else has predicted.

VII. Conclusions and Predictions

What has been established:

- $\sigma = 0.00323$ per e-fold (CMB reference) is the honest settling rate, derived from Paper XXIV without free parameters.
- $\varepsilon(z)$ evolves from ε_geom = 0.1688 at high-z toward ε_MSDS = 0.165 at z = 0, driven by dark energy growth.
- Dynamic $\varepsilon(z)$ predicts $\Delta\eta = -0.0022$ at z = 0.5, measurable by Euclid.
- The 14π ratio is a present-epoch fixed point, not a timeless invariant. It drifts significantly above z ~ 0.3.
- 14π emerges specifically under dark energy domination — linking it to the equation-of-state parameter w.

Open problem for Paper XXVI:

Why does dark energy domination produce the specific fixed point 14π? This requires deriving the ratio $\varepsilon_{geom}/\Delta\varepsilon$ analytically from the Ω_{DE}-tracking dynamics. If that derivation succeeds, the MSDS program will have predicted the dark energy density from geometry alone.

Paper XXIV found the blueprint. Paper XXV put it on a timeline. Paper XXVI must explain why the timeline ends at 14π.

References

[1] Planck Collaboration, A&A; 641 A6 (2020). Ω_DE,0 = 0.685, Ω_m,0 = 0.315.

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[5] Bellini, Sawicki, JCAP 2014(07) 050. Gravitational slip in Horndeski theories.

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